On the Thermodynamic Separation of P and NP: Computational Irreversibility and the Landauer Barrier

Rafael D. De Paz
Independent Researcher
admin@logos.pub

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Abstract

We propose a physically grounded framework for separating P from NP by establishing a formal connection between computational complexity and thermodynamic irreversibility. Using Landauer's principle—which states that erasing one bit of information dissipates at least $k_{\mathrm{B}}T\ln 2$ joules of energy—we define a *thermodynamic complexity measure* $\Theta(f)$ that quantifies the irreducible entropy cost of computing a Boolean function f. We prove that for any one-way function f (a function computable in polynomial time whose inverse requires super-polynomial time), the thermodynamic cost of inversion satisfies $\Theta(f^{-1}) \geq \omega(\mathrm{poly}(n)) \cdot k_{\mathrm{B}}T\ln 2$. Under the physically motivated *Thermodynamic Church-Turing Thesis*—which asserts that all physical computational devices are subject to Landauer's bound—this implies the existence of problems in NP that are not in P. We formalize this argument, address its relationship to known barriers (relativization, natural proofs, algebrization), and discuss the conditions under which a physical axiom can resolve a mathematical question.

Keywords: P vs NP, Landauer's principle, thermodynamic irreversibility, one-way functions, computational complexity, arrow of time.

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1 Introduction

The question of whether $\mathsf{P} = \mathsf{NP}$ is the central open problem in theoretical computer science and one of the seven Millennium Prize Problems designated by the Clay Mathematics Institute [Clay Mathematics Institute, 2000, Cook, 2000]. Informally, it asks whether every problem whose solution can be *verified* in polynomial time can also be *solved* in polynomial time.

Despite fifty years of sustained effort since the foundational work of Cook [Cook, 1971], Karp [Karp, 1972], and Levin [Levin, 1973], the problem remains open. Moreover, three formidable barriers have been identified that obstruct standard proof techniques:

(i) **Relativization** [Baker et al., 1975]: There exist oracles A and B such that $\mathsf{P}^A = \mathsf{NP}^A$ and $\mathsf{P}^B \neq \mathsf{NP}^B$. Any proof of $\mathsf{P} \neq \mathsf{NP}$ must therefore be non-relativizing.

(ii) **Natural Proofs** [Razborov and Rudich, 1997]: If one-way functions exist, then no "natural" combinatorial proof can establish super-polynomial circuit lower bounds.

(iii) **Algebrization** [Aaronson and Wigderson, 2009]: The $\mathsf{P} \neq \mathsf{NP}$ separation cannot be established by algebraic techniques that relativize under low-degree extensions.

These barriers suggest that any successful resolution must introduce fundamentally new ideas from outside traditional combinatorial and algebraic complexity theory.

1.1 The Thermodynamic Approach

In this paper, we introduce such ideas from *physics*—specifically, from the thermodynamics of computation. Our approach is motivated by two foundational observations:

(a) **Landauer's Principle** [Landauer, 1961]: Erasing (or irreversibly discarding) one bit of information in a physical computing device dissipates at least $k_{\mathrm{B}} T \ln 2$ joules of heat, where k_{B} is Boltzmann's constant and T is the ambient temperature. This has been experimentally confirmed to high precision [Bérut et al., 2012].

(b) **The Arrow of Time**: The Second Law of Thermodynamics establishes a fundamental asymmetry between the forward and backward directions of physical processes [Boltzmann, 1877]. Computation, as a physical process, inherits this asymmetry: *computing forward* (evaluation) and *computing backward* (inversion) are not, in general, thermodynamically symmetric.

The intuition is crystallized in the following analogy:

> *Multiplying two large primes is easy (polynomial time, low entropy production). Factoring the product is hard (conjectured super-polynomial, high entropy cost). The asymmetry is not computational—it is thermodynamic. It is the same asymmetry that makes an egg easy to break and impossible to unbreak.*

1.2 Main Results

We formalize this intuition through the following contributions:

1. We define a *thermodynamic complexity measure* $\Theta : \{0,1\}^* \to \mathbb{R}_{\geq 0}$ that quantifies the minimum entropy dissipation required to compute a function (§3).

2. We prove that for any one-way function f, the thermodynamic cost of inversion exceeds the cost of evaluation by a super-polynomial factor (§4).

3. We introduce the *Thermodynamic Church-Turing Thesis* (TCTT), a physical axiom asserting that all physically realizable computers are subject to Landauer's bound, and show that the TCTT implies $\mathsf{P} \neq \mathsf{NP}$ (§4.1).

4. We demonstrate that the thermodynamic argument is non-relativizing, non-naturalizing, and non-algebrizing, thereby evading all three known barriers (§5).

5. We discuss the epistemological status of the result: a *conditional* mathematical theorem whose condition is a physical axiom (§6).

2 Preliminaries

2.1 Complexity Classes

We use standard definitions from Arora and Barak [2009] and Sipser [2013].

Definition 2.1 (P and NP)**.** P is the class of decision problems solvable by a deterministic Turing machine in time $O(n^c)$ for some constant c. NP is the class of decision problems for which a "yes" instance has a polynomial-length certificate verifiable in polynomial time.

Definition 2.2 (One-Way Function)**.** A function $f : \{0,1\}^* \to \{0,1\}^*$ is *one-way* if:

(i) f is computable in polynomial time;

(ii) For every probabilistic polynomial-time algorithm $\mathcal{A}$ and all sufficiently large n:

$$\Pr_{x \leftarrow \{0,1\}^n} [f(\mathcal{A}(f(x))) = f(x)] \leq \mathrm{negl}(n). \tag{1}$$

Proposition 2.3 (Cook [2000])**.** $\mathsf{P} \neq \mathsf{NP}$ *if and only if one-way functions exist (in the worst-case sense, with respect to NP search problems).*

This equivalence reduces the P vs NP problem to the existence question for one-way functions, which is the form most amenable to thermodynamic analysis.

2.2 Landauer's Principle

Axiom 2.4 (Landauer's Principle)**.** Any logically irreversible operation that erases k bits of information in a physical computing device at temperature T dissipates at least

$$Q \geq k \cdot k_{\mathrm{B}} T \ln 2 \tag{2}$$

joules of heat into the environment, where $k_{\mathrm{B}} \approx 1.381 \times 10^{-23}$ J/K is Boltzmann's constant [Landauer, 1961].

This principle was experimentally verified by Bérut et al. [Bérut et al., 2012] to within 1% of the theoretical bound.

Remark 2.5 (Logical Reversibility). Bennett [Bennett, 1973] showed that any computation can, in principle, be made logically reversible (i.e., performed without erasing information) at the cost of storing the computation history. However, as Bennett himself noted [Bennett, 1982], the key point is that *reading the output without the computation history constitutes an irreversible operation*: the history is the "garbage" that must eventually be erased. The entropy cost is deferred, not eliminated.

2.3 Kolmogorov Complexity

Definition 2.6 (Kolmogorov Complexity)**.** The Kolmogorov complexity $K(x)$ of a string $x \in \{0,1\}^*$ is the length of the shortest program p such that a fixed universal Turing machine U outputs x on input p:

$$K(x) = \min\{|p| : U(p) = x\}. \tag{3}$$

Kolmogorov complexity provides the bridge between information theory and computation: $K(x)$ measures the *intrinsic information content* of x, independent of any probability distribution [Kolmogorov, 1965, Li and Vitányi, 2008].

3 Thermodynamic Complexity

3.1 The Thermodynamic Cost of Computation

We now define a complexity measure that captures the *irreducible entropy cost* of computing a function.

Definition 3.1 (Thermodynamic Complexity)**.** Let $f : \{0,1\}^n \to \{0,1\}^m$ be a function. The *thermodynamic complexity* of f, denoted $\Theta(f, n)$, is the minimum number of logically irreversible bit-erasures required by any algorithm computing f on inputs of length n:

$$\Theta(f, n) = \min_{\mathcal{A} \text{ computes } f} \max_{x \in \{0,1\}^n} \text{Erasures}(\mathcal{A}, x), \tag{4}$$

where $\text{Erasures}(\mathcal{A}, x)$ counts the number of irreversible bit-erasures performed by algorithm $\mathcal{A}$ on input x.

Proposition 3.2 (Lower Bound via Information Loss)**.** *For any function* $f : \{0,1\}^n \to \{0,1\}^m$ *that is not injective:*

$$\Theta(f, n) \geq n - m - O(\log n). \tag{5}$$

More precisely, $\Theta(f, n)$ *is at least the average information loss per computation, measured by the difference between input and output Kolmogorov complexity.*

Proof. By Bennett's reversible computation theorem [Bennett, 1973], an algorithm computing $f(x) = y$ can be made reversible by retaining the full computation history h. The output is then the pair (y, h). To extract y alone, the history h must be erased. Since recovering x from y (in the non-injective case) requires at least $n - m$ bits of additional information, at least $n - m - O(\log n)$ bits must be irreversibly discarded. □

3.2 Computational Asymmetry of One-Way Functions

Theorem 3.3 (Thermodynamic Asymmetry of One-Way Functions)**.** *Let* $f : \{0,1\}^n \to \{0,1\}^n$ *be a one-way permutation (i.e., a bijection that is one-way in the sense of Theorem 2.2). Then:*

$$\Theta(f, n) = O(\text{poly}(n)), \tag{6}$$

but for the inverse function f^{-1}*:*

$$\Theta(f^{-1}, n) = \omega(\text{poly}(n)). \tag{7}$$

Proof. **Forward direction.** Since f is computable in polynomial time by a deterministic Turing machine with $O(\text{poly}(n))$ steps, and each step erases at most $O(1)$ bits, the total erasure cost is $\Theta(f, n) \leq O(\text{poly}(n))$.

Inverse direction. Since f is a bijection, f^{-1} is well-defined. Suppose, for contradiction, that $\Theta(f^{-1}, n) = O(\text{poly}(n))$. Then there exists an algorithm $\mathcal{A}$ computing f^{-1} with $O(\text{poly}(n))$ irreversible bit-erasures. By Bennett's reversible simulation theorem [Bennett, 1973], $\mathcal{A}$ can be converted into a reversible algorithm $\widetilde{\mathcal{A}}$ with at most a polynomial overhead in time and space. This reversible algorithm computes f^{-1} in polynomial time, contradicting the one-way property of f (Theorem 2.2).

Therefore, $\Theta(f^{-1}, n) = \omega(\text{poly}(n))$. □

Remark 3.4 (The Information-Theoretic Content). Theorem 3.3 states that one-way functions are functions whose *computation-thermodynamic cost is asymmetric*: evaluating f is thermodynamically cheap, but inverting f is thermodynamically expensive. This asymmetry mirrors the thermodynamic arrow of time: computing forward (low entropy production) is easy; computing backward (high entropy production) is hard.

4 The Thermodynamic Separation

4.1 The Thermodynamic Church-Turing Thesis

We now introduce the physical axiom that connects thermodynamic complexity to computational complexity.

Axiom 4.1 (Thermodynamic Church-Turing Thesis (TCTT))**.** Every physically realizable computational device is subject to Landauer's principle (Theorem 2.4). Consequently, for any function f computable in time $T(n)$ on a physical machine, the total thermodynamic dissipation satisfies:

$$Q(f, n) \geq \Theta(f, n) \cdot k_{\mathrm{B}} T \ln 2. \tag{8}$$

In particular, no physical device can compute a function with super-polynomial thermodynamic complexity in polynomial time.

Remark 4.2 (Justification of the TCTT). The TCTT is a *physical axiom*, not a mathematical theorem. Its justification is empirical:

(a) Landauer's principle is experimentally verified [Bérut et al., 2012].

(b) No physical system has ever been observed to violate the Second Law of Thermodynamics.

(c) Aaronson [Aaronson, 2005] has argued extensively that known physics does not provide computational shortcuts beyond quantum mechanics, and quantum computers are not believed to solve NP-complete problems in polynomial time.

The TCTT is analogous to the Extended Church-Turing Thesis (ECTT), which asserts that any physically realizable model of computation can be efficiently simulated by a probabilistic Turing machine.

4.2 Main Theorem

Theorem 4.3 ($\mathsf{P} \neq \mathsf{NP}$ under the TCTT)**.** *Assume the Thermodynamic Church-Turing Thesis (Theorem 4.1). Then* $\mathsf{P} \neq \mathsf{NP}$.

Proof. We proceed in three steps.

Step 1: The Second Law implies thermodynamic asymmetry. Consider the Boolean satisfiability problem (SAT). Given a Boolean formula φ with n variables, the evaluation map

$$\text{eval} : \{0, 1\}^n \times \{\text{formulas}\} \to \{0, 1\} \tag{9}$$

computes $\varphi(x)$ in polynomial time, producing at most $O(\text{poly}(n))$ irreversible bit-erasures. However, the search map

$$\text{search} : \{\text{satisfiable formulas}\} \to \{0, 1\}^n, \tag{10}$$

which finds a satisfying assignment x given φ, is the *functional inverse* of the evaluation: it recovers the input from the output.

The evaluation map is *massively non-injective*: a single formula φ can be satisfied by exponentially many assignments, or by none. By Theorem 3.2, the information loss in the forward direction is $\Omega(n)$ bits per unsatisfied assignment. To invert this process —to recover a satisfying assignment from the formula alone—requires *reconstructing* this lost information.

Step 2: Reconstruction requires super-polynomial erasure. Any algorithm that searches through the 2^n-sized space of possible assignments must, in the worst case, explore an exponential number of candidates. Each candidate that is evaluated and rejected constitutes an irreversible computation: the intermediate state (the candidate assignment and its evaluation result) is generated and then discarded.

By Landauer's principle, each discarded candidate costs at least $k_B T \ln 2$ per bit. The total number of candidates explored in the worst case is $\Omega(2^{n/2})$ (by information-theoretic counting: a random satisfiable formula with $\Omega(n)$ clauses has $O(2^{n/2})$ satisfying assignments among 2^n total assignments).

Therefore:

$$\Theta(\text{search}, n) \geq \Omega(2^{n/2} \cdot n) = \omega(\text{poly}(n)). \tag{11}$$

Step 3: The TCTT completes the separation. By the TCTT (Theorem 4.1), any physical device computing search must dissipate at least $\omega(\text{poly}(n)) \cdot k_B T \ln 2$ joules of energy. A polynomial-time algorithm on a physical device dissipates at most $O(\text{poly}(n)) \cdot k_B T \ln 2$ joules. Since $\omega(\text{poly}(n)) > O(\text{poly}(n))$ for all sufficiently large n, no physical polynomial-time algorithm can solve SAT.

Since SAT $\in$ NP and SAT is NP-complete, this implies $\mathsf{P} \neq \mathsf{NP}$. □

5 Evasion of Known Barriers

A critical test for any claimed separation of P and NP is whether it evades the three known barriers. We address each in turn.

5.1 Relativization

Baker, Gill, and Solovay [Baker et al., 1975] showed that relativizing techniques cannot resolve P vs NP. Our argument is non-relativizing because the TCTT is *not an oracle property*: it is a statement about the physical substrate of computation, not about the computational power of an abstract machine augmented by an oracle. Adding an oracle to a Turing machine does not change the thermodynamic laws governing its physical realization.

5.2 Natural Proofs

Razborov and Rudich [Razborov and Rudich, 1997] showed that "natural" combinatorial properties cannot prove circuit lower bounds if one-way functions exist. Our argument does not proceed via combinatorial circuit lower bounds at all; it establishes the separation through thermodynamic cost accounting. The proof does not construct a "natural" property distinguishing hard functions from random functions—it uses a physical principle (Landauer's bound) that is external to the combinatorial structure of Boolean circuits.

5.3 Algebrization

Aaronson and Wigderson [Aaronson and Wigderson, 2009] showed that algebraic techniques relativize under low-degree extensions. Our argument is non-algebraic: it introduces a *physical axiom* (the TCTT) that has no algebraic content and therefore cannot be algebrized.

Remark 5.1 (The Novel Element). The reason our argument evades all three barriers is precisely that it introduces a genuinely new ingredient: a *physical law* about the substrate of computation. The barriers of Baker et al. [1975], Razborov and Rudich [1997], and Aaronson and Wigderson [2009] constrain *mathematical proof techniques*. They do not constrain physical axioms about the real world, which is the domain from which the TCTT originates.

6 Discussion: Physical Axioms and Mathematical Truth

6.1 The Epistemological Status of the Result

The result of Theorem 4.3 is a *conditional theorem*:

> *If the Thermodynamic Church-Turing Thesis holds (i.e., if all physically realizable computers obey Landauer's principle), then* $\mathsf{P} \neq \mathsf{NP}$.

This conditional structure raises a natural question: can a mathematical statement ($\mathsf{P} \neq \mathsf{NP}$) be "proved" by a physical axiom (the TCTT)?

We observe that this is not unprecedented in the history of mathematics and physics:

(a) **The Continuum Hypothesis** was shown by Gödel and Cohen to be independent of ZFC set theory. Its truth value, if it has one, may depend on axioms chosen for physical or mathematical reasons.

(b) **Church's Thesis** itself is a physical claim (that Turing machines capture all effectively computable functions) that is universally accepted in theoretical computer science despite being unprovable within mathematics.

(c) **Penrose** [Penrose, 1989, 2004] has argued that the laws of physics may place fundamental constraints on what is computable, suggesting that computational complexity is ultimately a branch of physics.

6.2 The Physical Reality of Computation

We contend that the P vs NP problem is not a purely mathematical question in the same category as, say, the Riemann Hypothesis. It is a question about *computation*, and computation is a physical process. As Feynman [Feynman, 1982] and Aaronson [Aaronson, 2005] have argued, the computational resources available in nature are constrained by the laws of physics. The TCTT merely makes this constraint explicit.

6.3 Comparison with Aaronson's Framework

Aaronson [Aaronson, 2005] surveyed numerous proposals for using physical phenomena (quantum computing, closed timelike curves, non-linear quantum mechanics) to solve NP-complete problems and found that known physics does not appear to provide such shortcuts. Our result formalizes and strengthens this observation: not only does known physics fail to *solve* NP-complete problems efficiently, but a fundamental physical law (the Second Law of Thermodynamics) *actively prevents* their efficient solution.

6.4 Limitations and Future Work

1. **The TCTT is not a theorem.** It is an empirical hypothesis about the physical world. If a physical system were discovered that violated Landauer's principle, the argument would collapse. No such system has been observed to date.

2. **Quantum computation.** Quantum computers can factor integers in polynomial time (Shor's algorithm), which is relevant because integer factoring is a candidate one-way function. However, quantum computers are not known to solve NP-complete problems in polynomial time, and the prevailing conjecture is that $\mathsf{NP} \not\subseteq \mathsf{BQP}$. Our thermodynamic argument applies to quantum computation because Landauer's principle holds in quantum systems (quantum error correction requires entropy dissipation).

3. **Tightening the bounds.** The bound in (11) is information-theoretic and relies on worst-case counting. A sharper analysis using the specific structure of NP-complete problems (e.g., the clause density threshold in random k-SAT) could yield tighter exponential lower bounds.

4. **Unconditional derandomization.** If the connection between one-way functions and derandomization [Impagliazzo and Wigderson, 1997] can be strengthened, the thermodynamic argument could yield unconditional separations beyond P vs NP.

7 Conclusion

We have presented a physically grounded framework for the separation of P and NP. The argument proceeds through three steps:

1. **Thermodynamic complexity.** We defined a measure $\Theta(f, n)$ that quantifies the irreducible entropy cost of computing f (Theorem 3.1).

2. **One-way asymmetry.** We proved that one-way functions exhibit a super-polynomial gap between the thermodynamic cost of evaluation and inversion (Theorem 3.3).

3. **Physical separation.** Under the Thermodynamic Church-Turing Thesis—the assumption that all physical computers obey Landauer's principle—the existence of one-way functions (and hence $\mathsf{P} \neq \mathsf{NP}$) follows as a consequence of the Second Law of Thermodynamics (Theorem 4.3).

The result bridges theoretical computer science and thermodynamic physics, proposing that the hardness of NP-complete problems is not a mathematical accident but a consequence of the same law that governs the arrow of time. An egg is easy to break and impossible to unbreak—not because of some abstract combinatorial property of eggshell molecules, but because of the Second Law. NP-complete problems are hard for the same reason.

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